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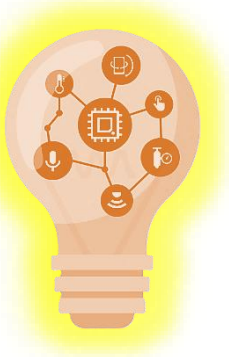

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# Statistis - Basics

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Ref: Web, Slides UI through

# intro

Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting data to assist in making more effective decisions.

Statistics presents a rigorous scientific method for gaining insight into data.

For example, suppose we measure the weight of 100 patients in a study. With so many measurements, simply looking at the data fails to provide an informative account.
However, statistics can give an instant overall picture of data based on graphical presentation or numerical summarization, irrespective to the number of data points.
Besides data summarization, another important task of statistics is to make inferences and predict relations between variables

# statistical analysis

**Statistical analysis** – used to manipulate, summarize, and investigate data, so that useful decision-making information results.

**Descriptive** statistics:
Methods of organizing, summarizing, and presenting data in an informative way

**Inferential** statistics:
The methods used to determine something about a population based on a sample

**Population:**
The entire set of individuals or objects of interest, or the measurements obtained from all individuals or objects of interest

**Sample:**
A portion, or part, of the population of interest

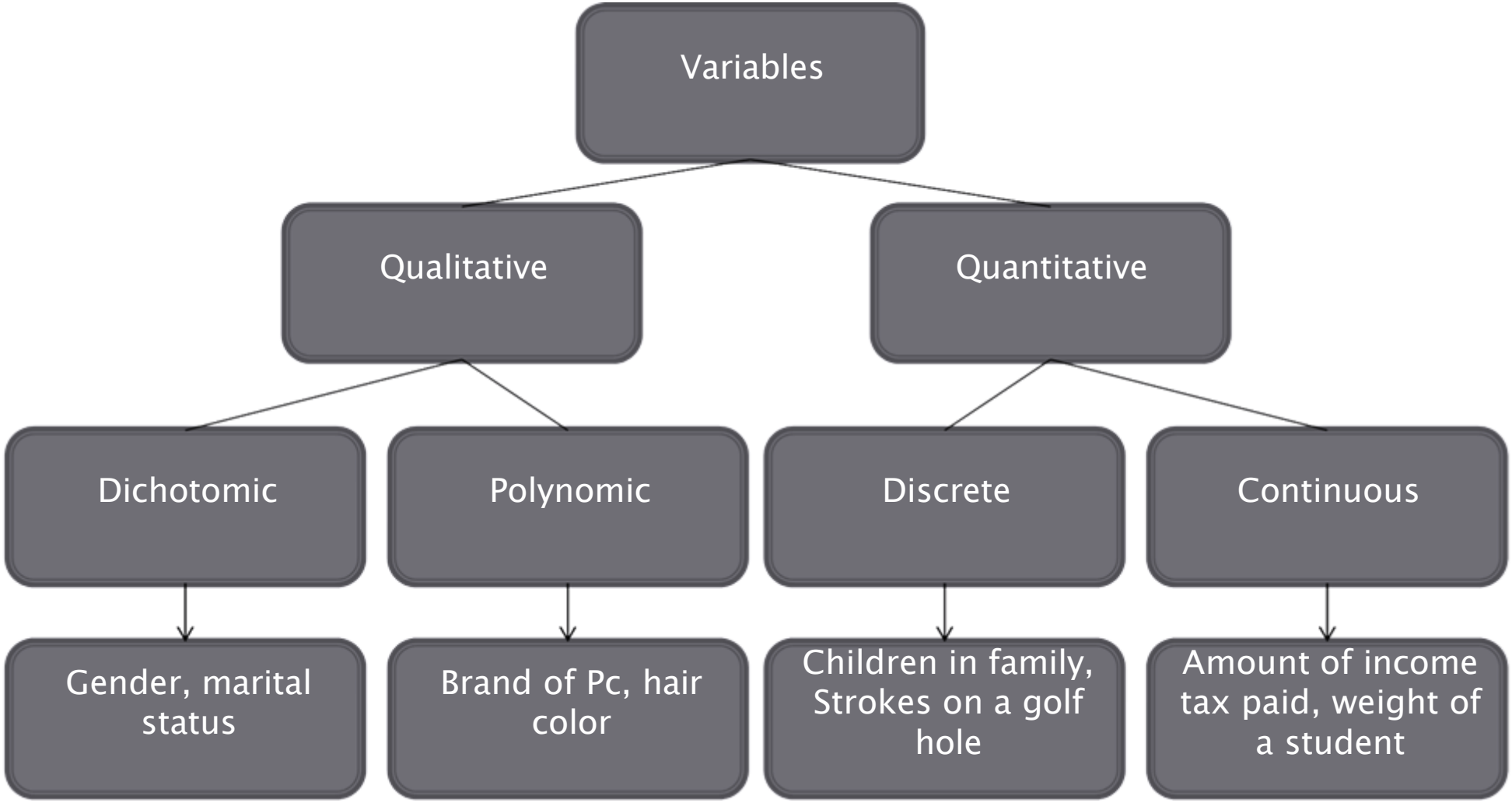
# statistical data

The collection of data that is relevant to the problem being studied is commonly the most difficult, expensive, and time-consuming part of the entire research project.

Statistical data are usually obtained by counting or measuring items.
Primary data are collected specifically for the analysis desired
Secondary data have already been compiled and are available for statistical analysis
A variable is an item of interest that can take on many different numerical values.
A constant has a fixed numerical value.

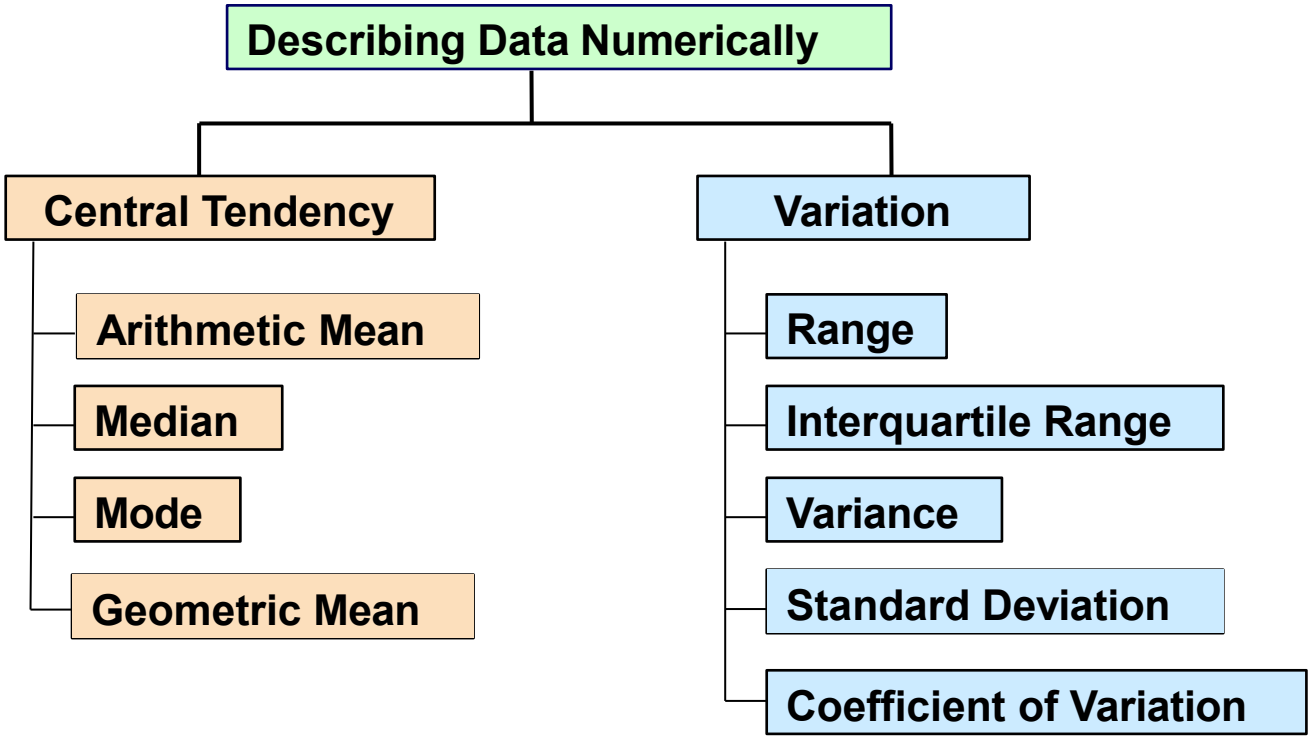
Statistical data are usually obtained by counting or measuring items. Most data can be put into the following categories:
**Qualitative** data can be separated into two subgroups:
dichotomic (if it takes the form of a word with two options (gender - male or female)
polynomic (if it takes the form of a word with more than two options (education - primary school, secondary school, and university).
**Quantitative** data are always numbers and result from counting or measuring attributes of a population. Quantitative data can be separated into two subgroups:
discrete (if it is the result of counting (the number of students of a given ethnic group in a class, the number of books on a shelf, ...)
continuous (if it is the result of measuring (distance traveled, weight of luggage, ...)

# types of variables



# why descriptive statistics?

- Analysts always need to become acquainted with numerical descriptive measures that provide very brief and easy-to-understand summaries of a data collection
- There are two broad categories into which these measures fail:
  - measures of central tendency and
  - measures of variability



# central tendency

**Center measurement** is a summary measure of the overall level of a dataset

Outlier: An individual value that falls outside theoverall pattern is called an outlier.

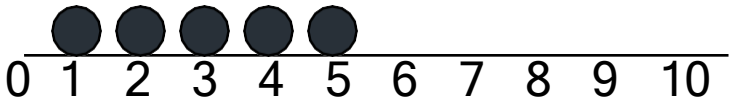
Mean = sum of values divided by the number of values

Affected by extreme values (outliers)

$$\bar{x} = \frac{\sum_{i=1} x_i}{n} \text{ (simple) or } \bar{x} = \frac{\sum_{i=1} x_i \cdot n_i}{n} \text{ (weighted)}$$

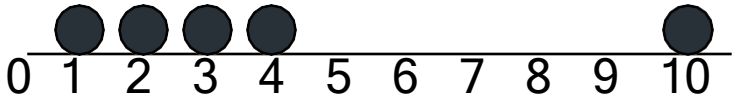
Observed values

Sample size




$$\frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

Mean = 3




$$\frac{1+2+3+4+10}{5} = \frac{20}{5} = 4$$

Mean = 4

$$Median\ position = \frac{n+1}{2}\ position\ in\ the\ ordered\ data$$

NOT affected by outliers

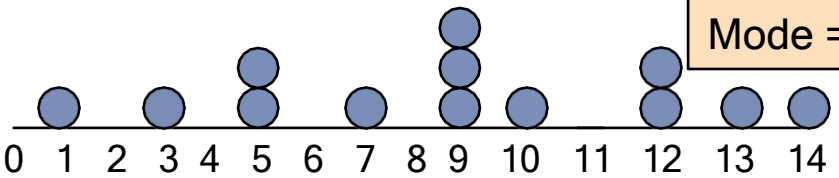
Median = 3

Median = 3

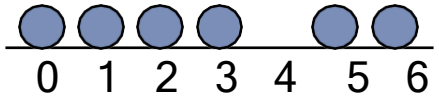
Mode: Value that occurs most often

NOT affected by outliers

Mode = 9




No Mode



# variance

The variance of a set of observations is the average of the squares of the deviations of the observations from their mean.

## Sample variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \text{ (simple) or } s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot n_i}{n-1} \text{ (weighted)}$$

where $\bar{x}$ = arithmetic mean n = sample size

$x_i$ = $i^{th}$ value of the variable x

$n_i$ = absolute frequency

# standard deviation

- The square root of the sample variance
- Most commonly used measure of variation
- Shows variation about the mean
- Has the same units as the original data

## Sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} \text{ (simple) or } s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot n_i}{n-1}} \text{ (weighted)}$$

# example: standard deviation

Sample Data ($x_i$) :

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 10 | 12 | 14 | 15 | 17 | 18 | 18 | 24 |

n = 8 Mean = $\bar{x}$ = 16

$$s = \sqrt{\frac{(10-\bar{X})^2 + (12-\bar{x})^2 + (14-\bar{x})^2 + \square + (24-\bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{(10-16)^2 + (12-16)^2 + (14-16)^2 + \square + (24-16)^2}{8-1}}$$

$$= \sqrt{\frac{126}{7}} = \boxed{4.2426}$$

A measure of the "average" scatter around the mean

# coefficient of variation

- Measures relative variation
- Always in percentage (%)
- Shows variation relative to mean
- Can be used to compare two or more sets data of measured in different units

$$cv = \left( \frac{s}{\bar{x}} \right) \cdot 100\%$$

# example

- Stock A:
  - Average price last year = $50
  - Standard deviation = $5

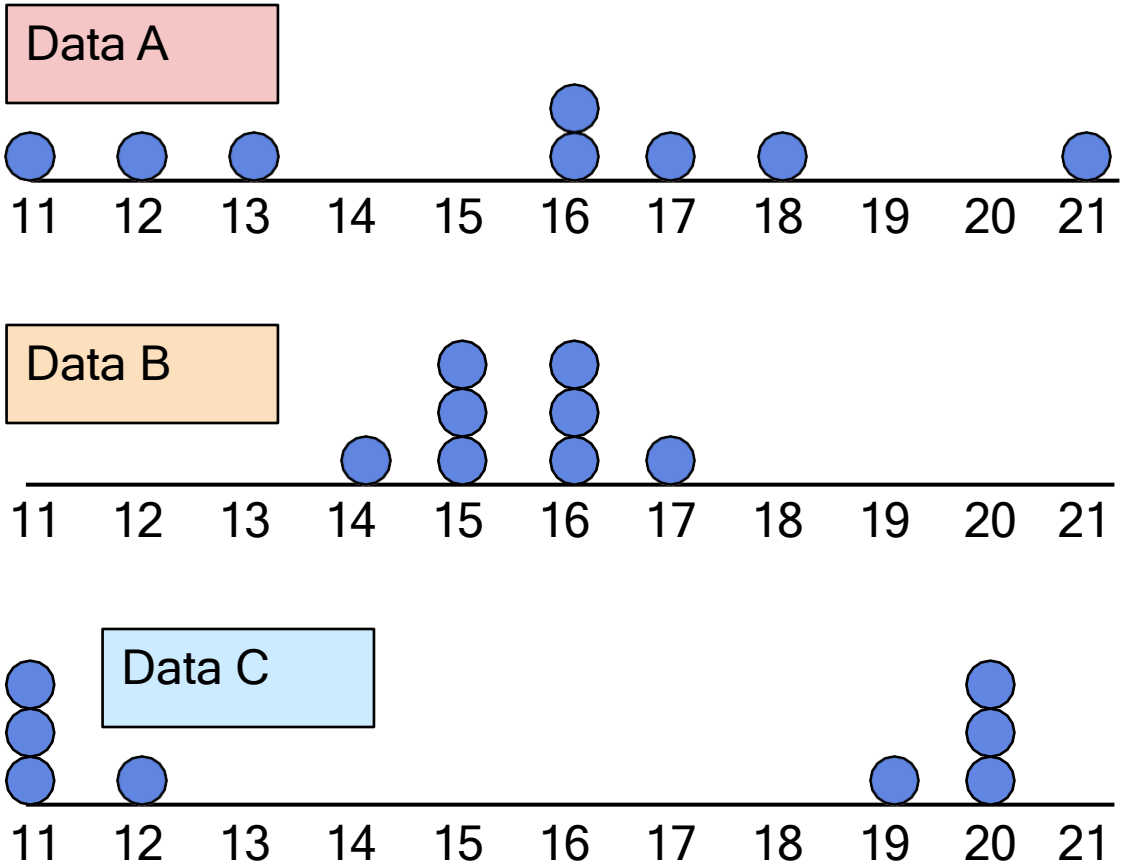
$$cv_A = \left(\frac{s}{\bar{x}}\right)\cdot 100\% = \frac{\$5}{\$50}\cdot 100\% = 10\%$$

- Stock B:
  - Average price last year = $100
  - Standard deviation = $5

$$cv_B = \left(\frac{s}{\bar{x}}\right)\cdot 100\% = \frac{\$5}{\$100}\cdot 100\% = 5\%$$

Both stocks have the same standard deviation, but stock B is less variable relative to its price

# comparing standard deviations



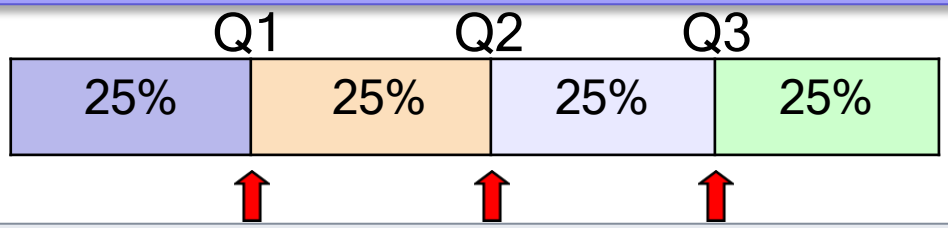

Mean = 15.5
s = 3.338

Mean = 15.5
s = 0.926

Mean = 15.5
s = 4.570

# quantiles

Quartiles split the ranked data into 4 segments with an equal number of values per segment




- The first quartile, $Q_1$, is the value for which 25% of the observations are smaller and 75% are larger
- $Q_2$ is the same as the median (50% are smaller, 50% are larger)
- Only 25% of the observations are greater than the third quartile

Find a quartile by determining the value in the appropriate position in the ranked data, where

First quartile position: $Q_1 = 0.25(n+1)$

Second quartile position: $Q_2 = 0.50(n+1)$

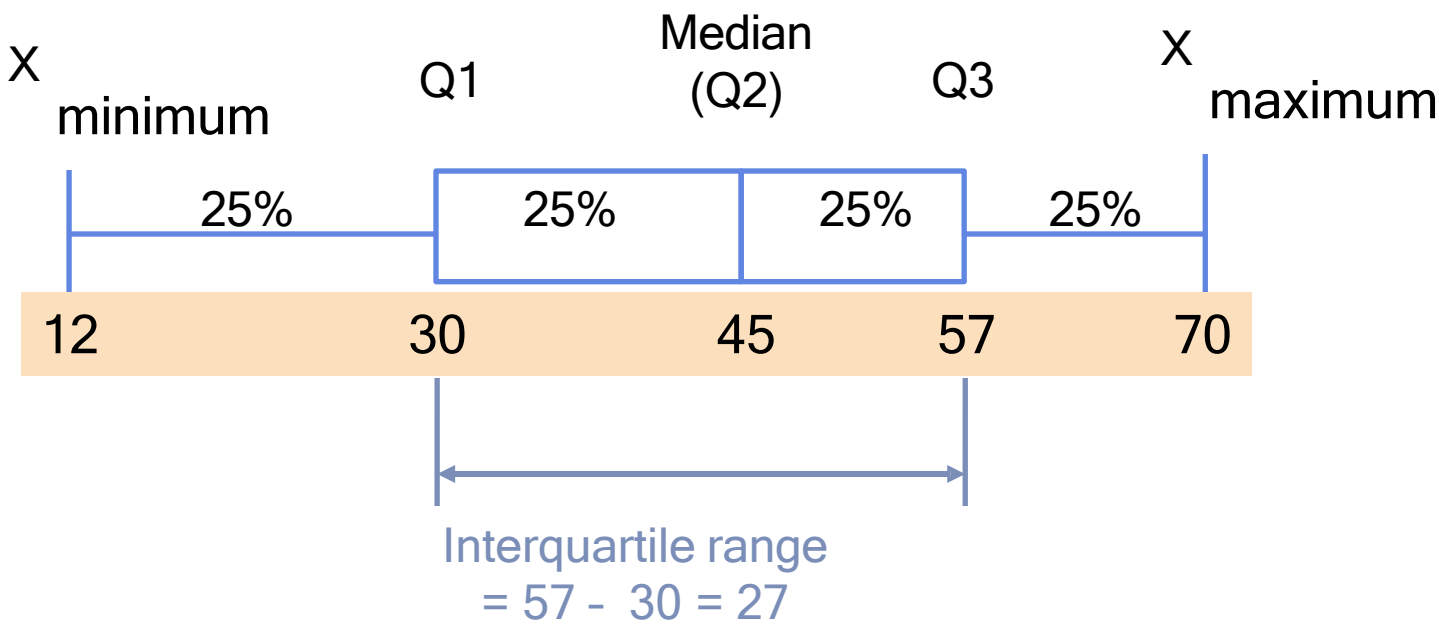
(the median position)

Third quartile position: $Q_3 = 0.75(n+1)$

where n is the number of observed values

# interquartile range - box plot

Example:




Interquartile range = 3rd quartile - 1st quartile

IQR = $Q_3$ - $Q_1$